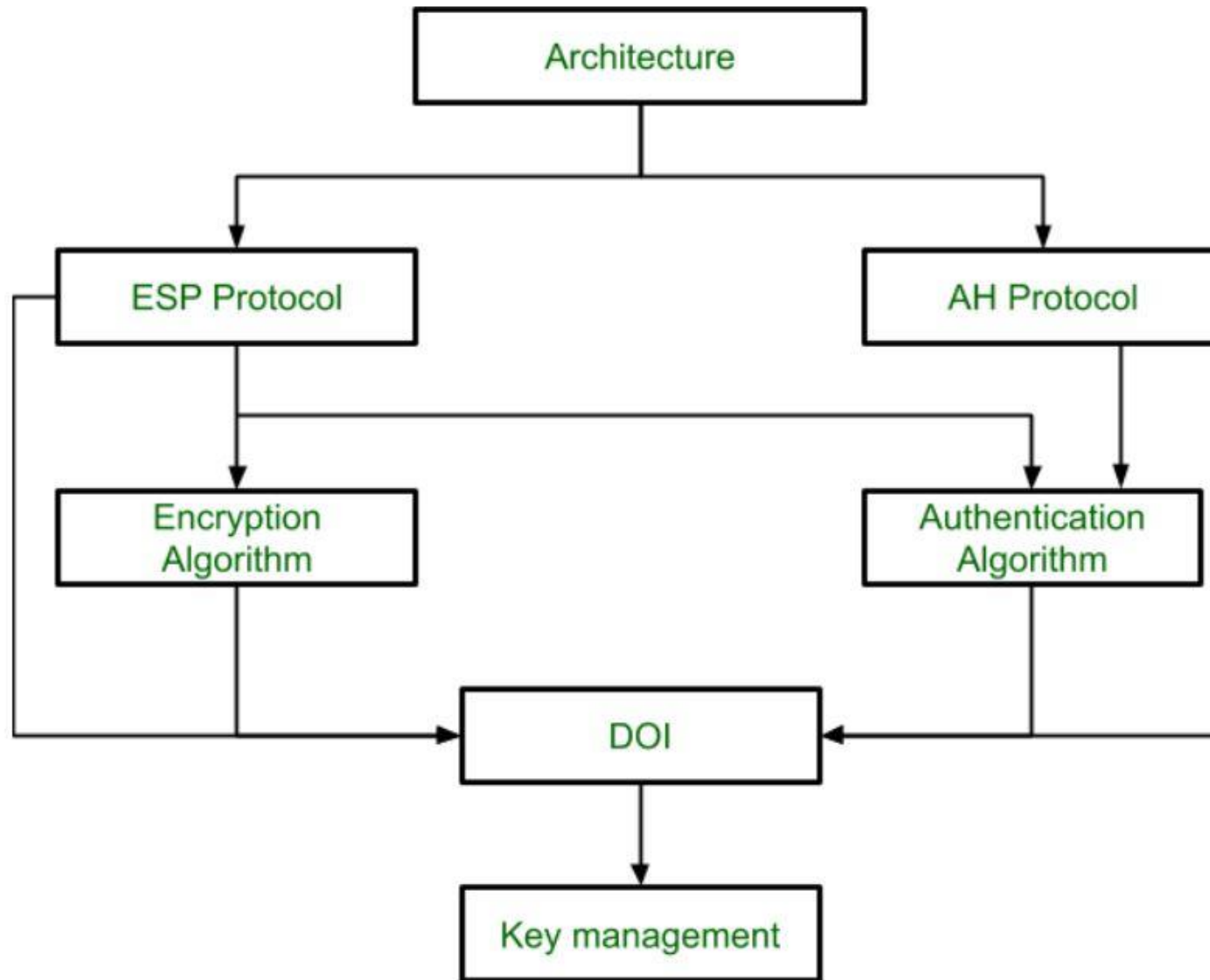


Network Security

IPSec (IP Security) Architecture

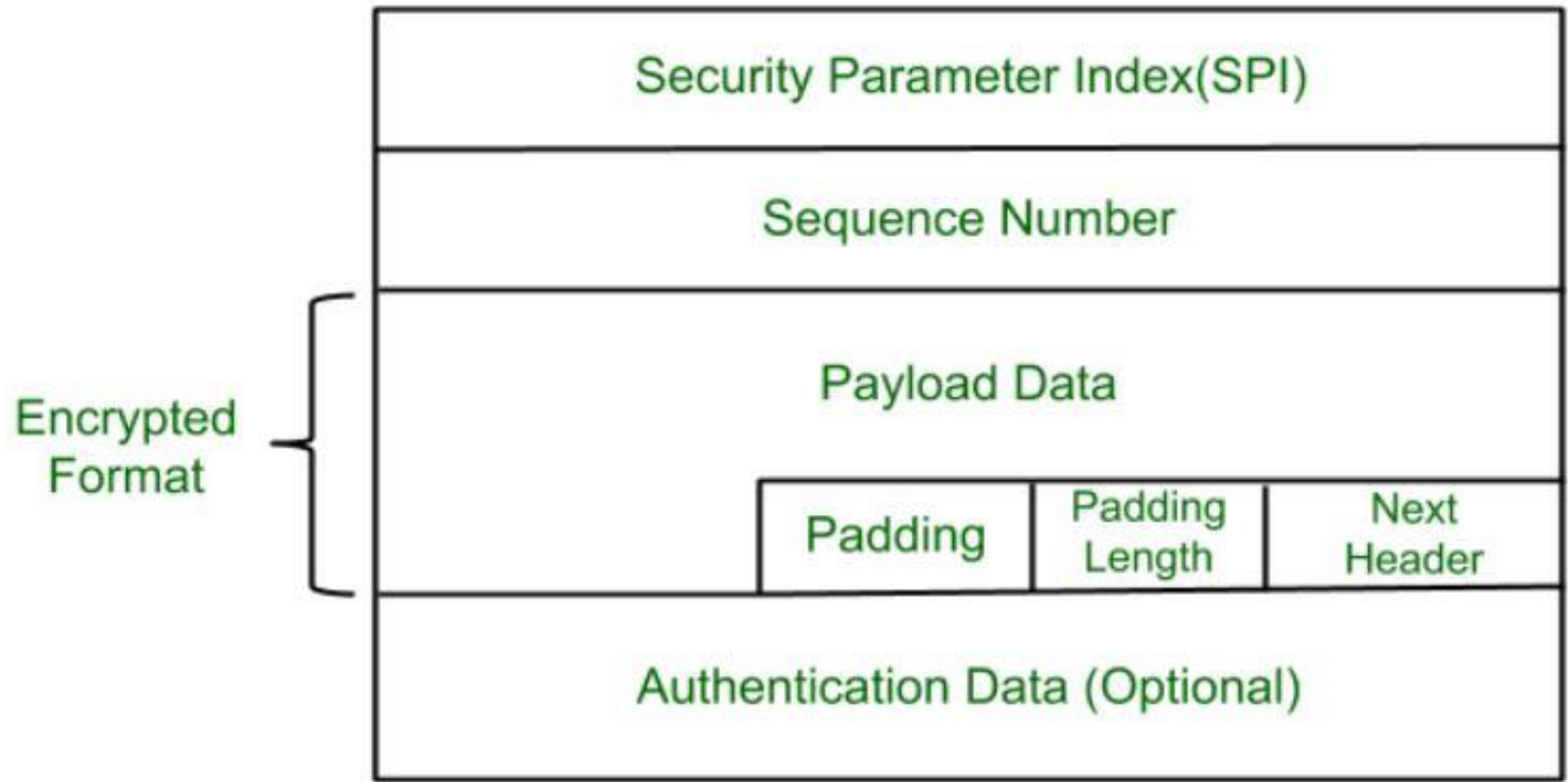
- **IPSec (IP Security) architecture** uses two protocols to secure the traffic or data flow.
- These protocols are **ESP (Encapsulation Security Payload) and AH (Authentication Header)**.
- IPSec Architecture includes
 1. Protocols,
 2. Algorithms,
 3. DOI, and
 4. Key Management.
- All these components are very important in order to provide the three main services:
 1. Confidentiality
 2. Authentication
 3. Integrity



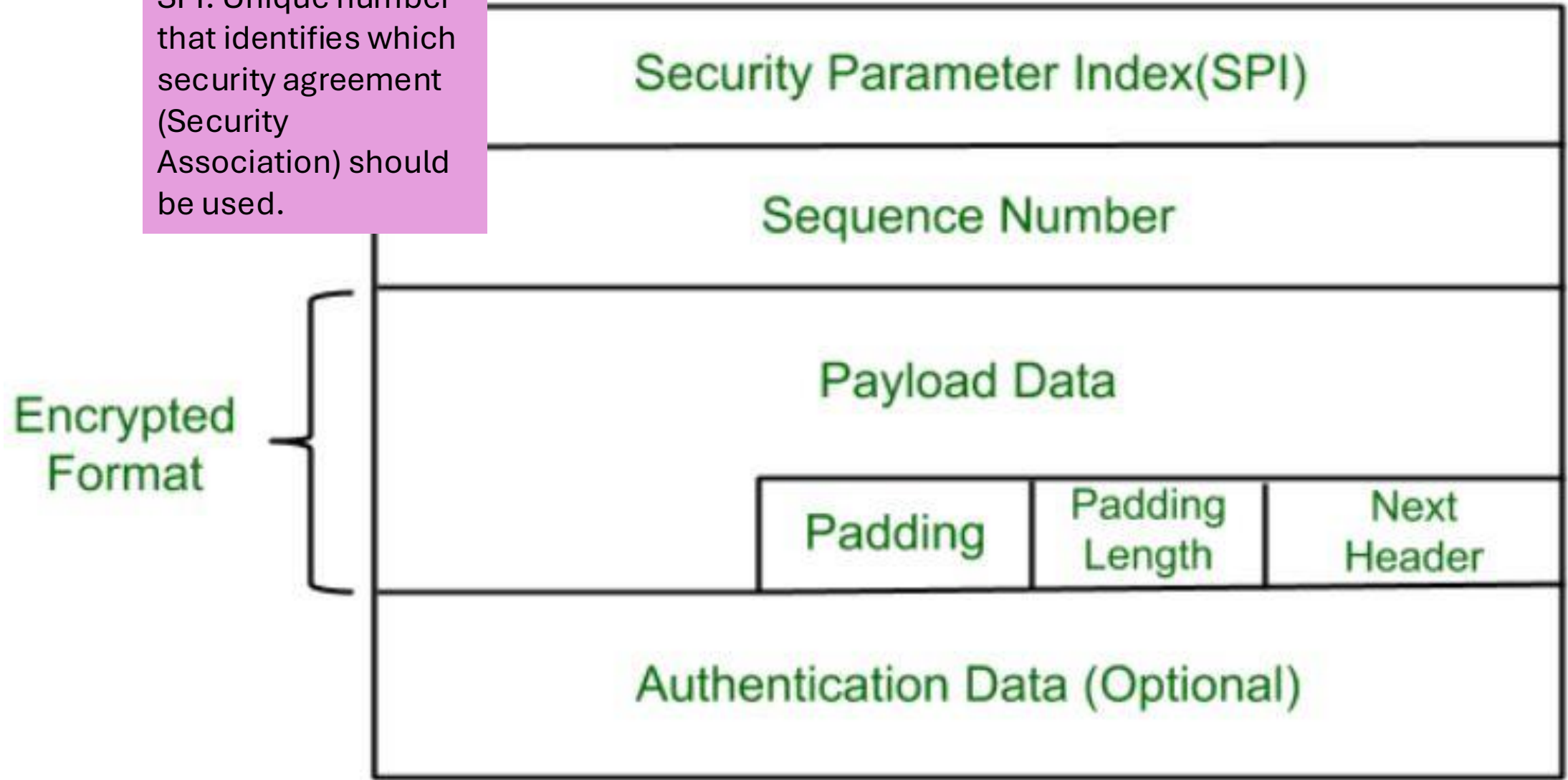
ESP Protocol

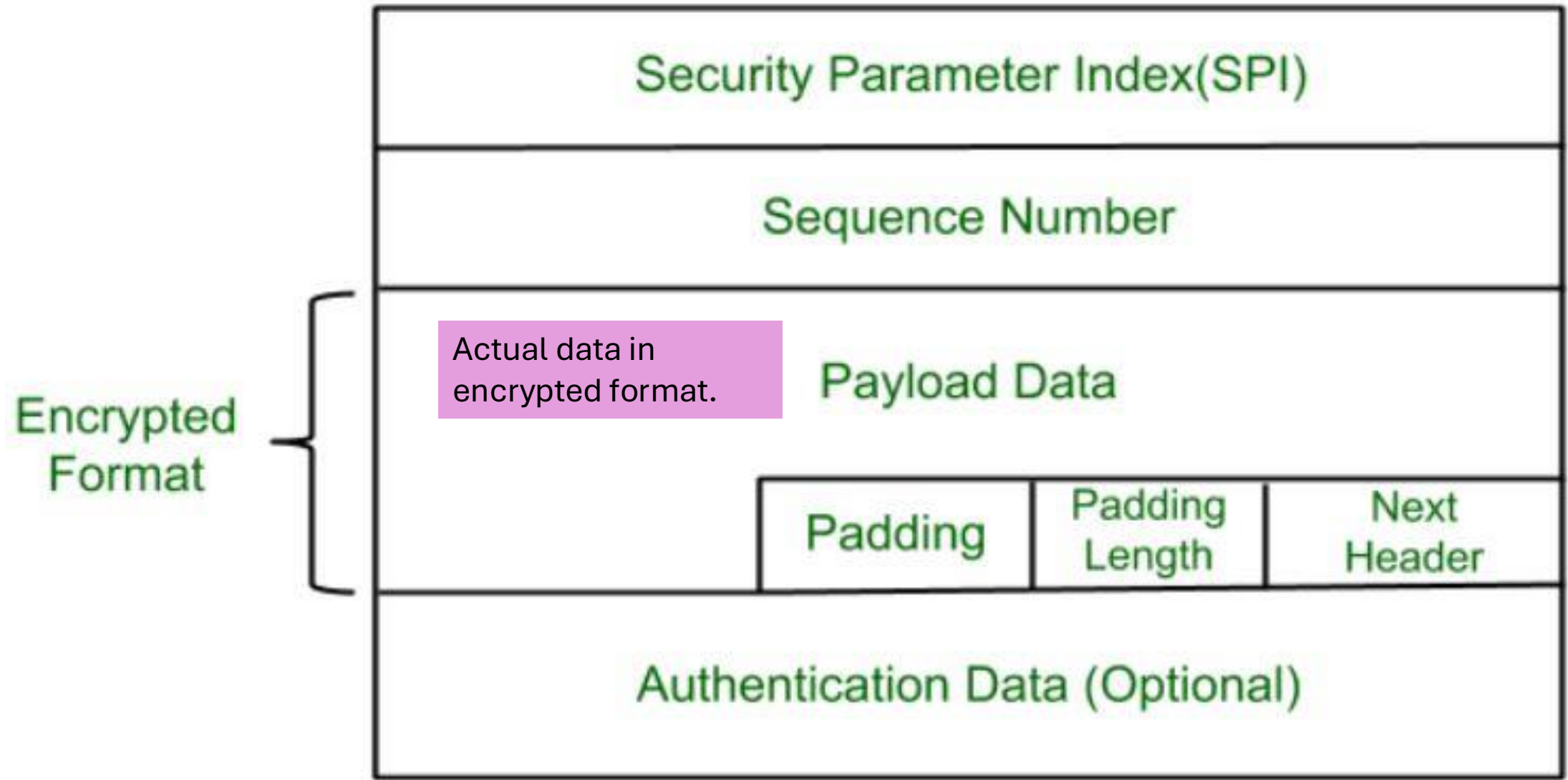
Encapsulation Security Payload (ESP) protocol

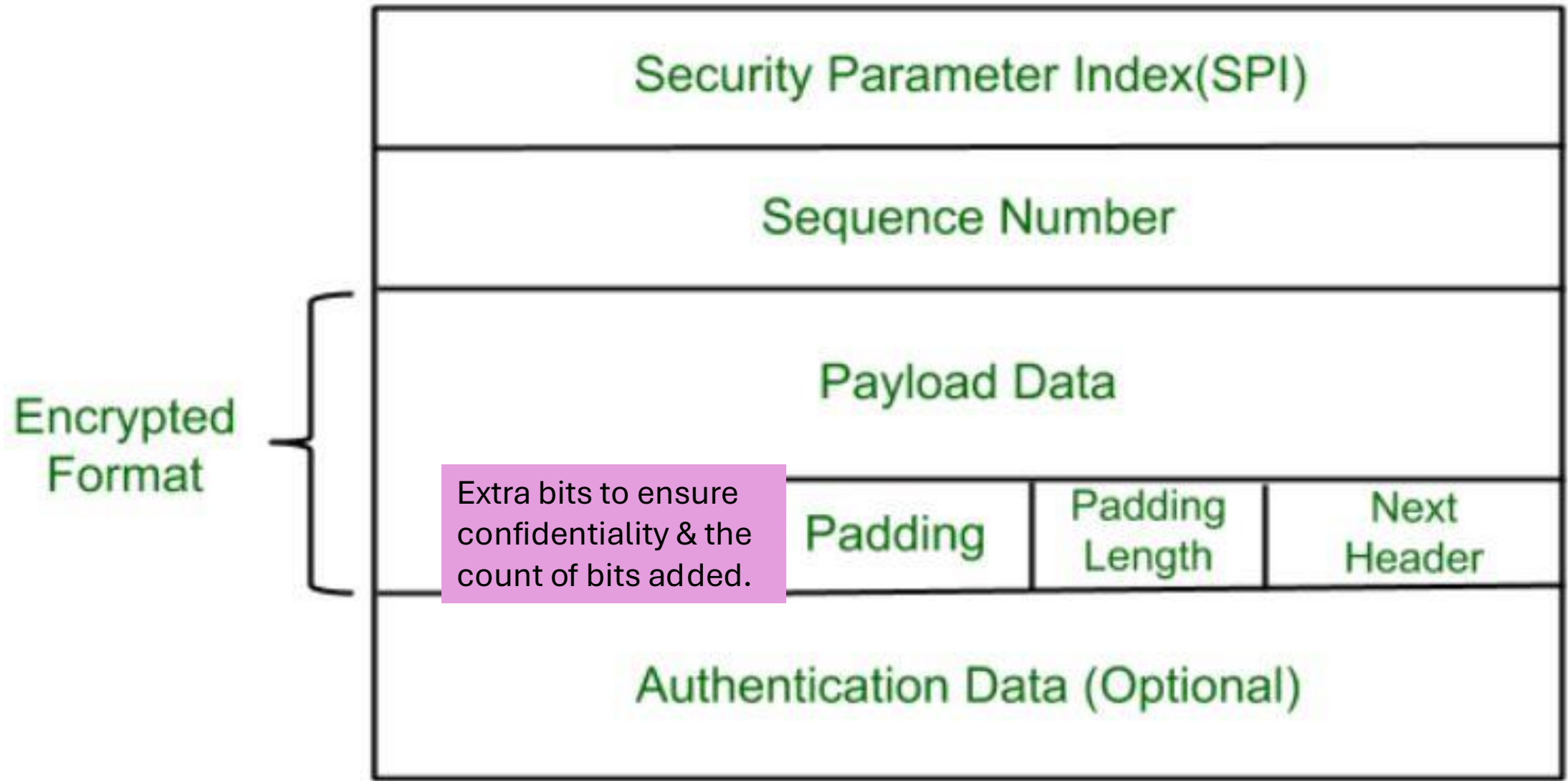
- ESP provides a confidentiality service.
- Encapsulation Security Payload is implemented in either two ways:
 1. ESP with optional Authentication.
 2. ESP with Authentication.

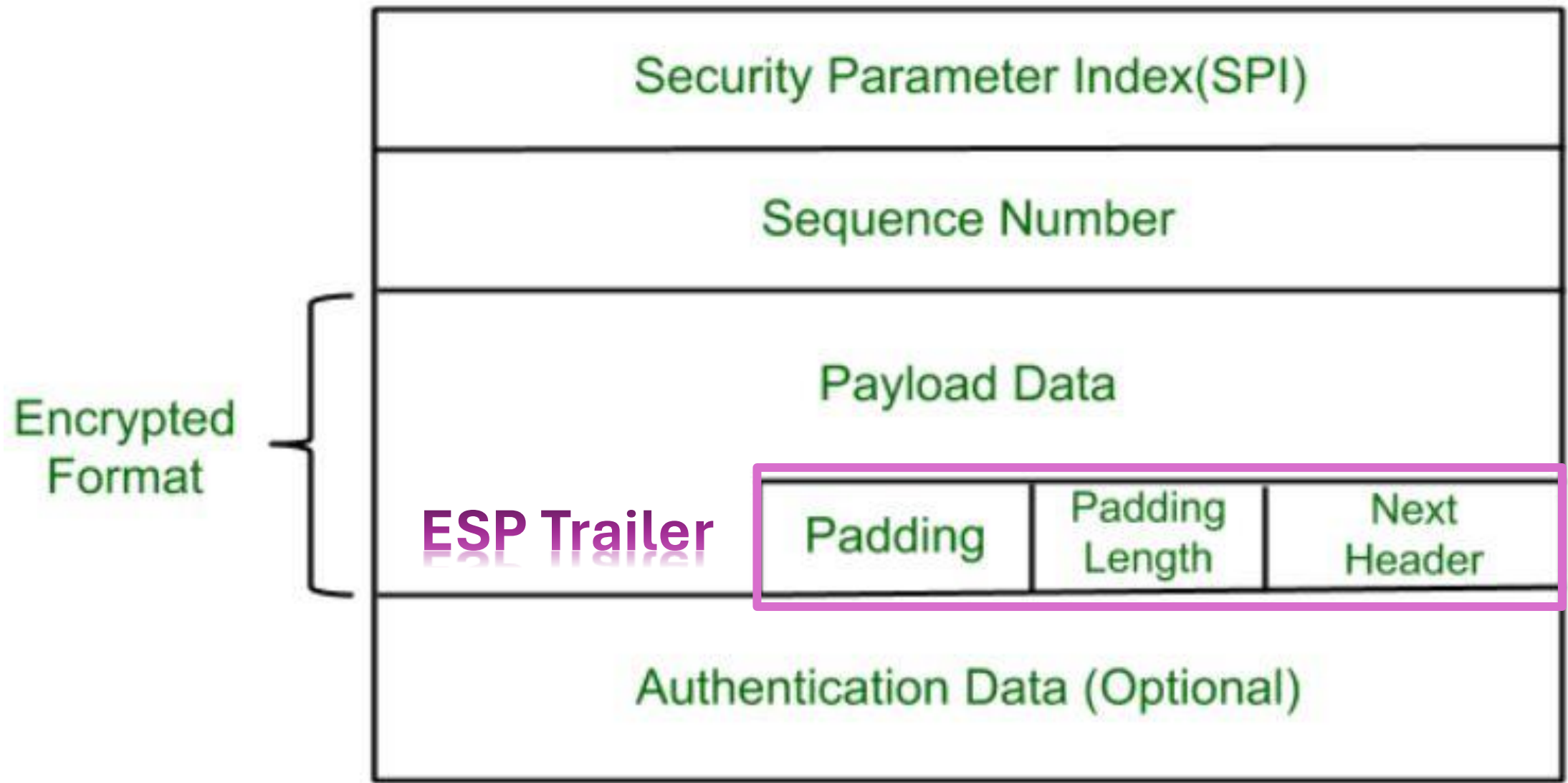


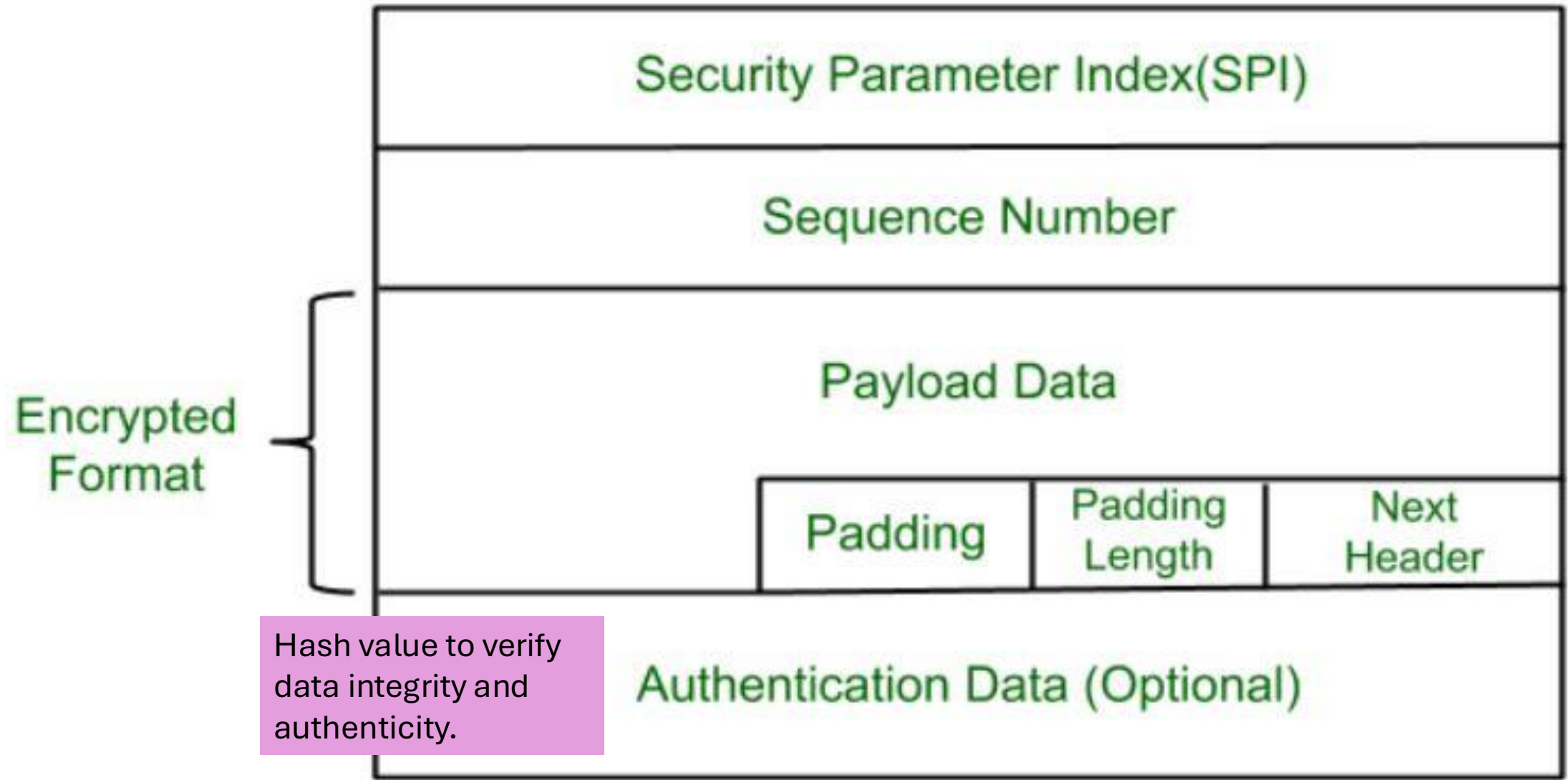
SPI: Unique number that identifies which security agreement (Security Association) should be used.











- **Security Parameter Index(SPI):** This parameter is used by Security Association. It is used to give a unique number to the connection built between the Client and Server.
- **Sequence Number:** Unique Sequence numbers are allotted to every packet so that on the receiver side packets can be arranged properly.
- **Payload Data:** Payload data means the actual data or the actual message. The Payload data is in an encrypted format to achieve confidentiality.
- **Padding:** Extra bits of space are added to the original message in order to ensure confidentiality. Padding length is the size of the added bits of space in the original message.
- **Next Header:** Next header means the next payload or next actual data.
- **Authentication Data** Contains a hash value used to verify data integrity and authenticity, ensuring the packet has not been modified during transmission.

Modes in ESP:

1. Tunnel Mode:

- In this mode, the entire original IP packet (header + payload) is encrypted and encapsulated inside a new packet with a new outer IP header, providing higher security.
- A new outer IP header is created to hide the original sender and receiver details.
- Provides complete protection to both payload and original IP header.
- Commonly used in **VPNs and gateway-to-gateway communication** for strong security.

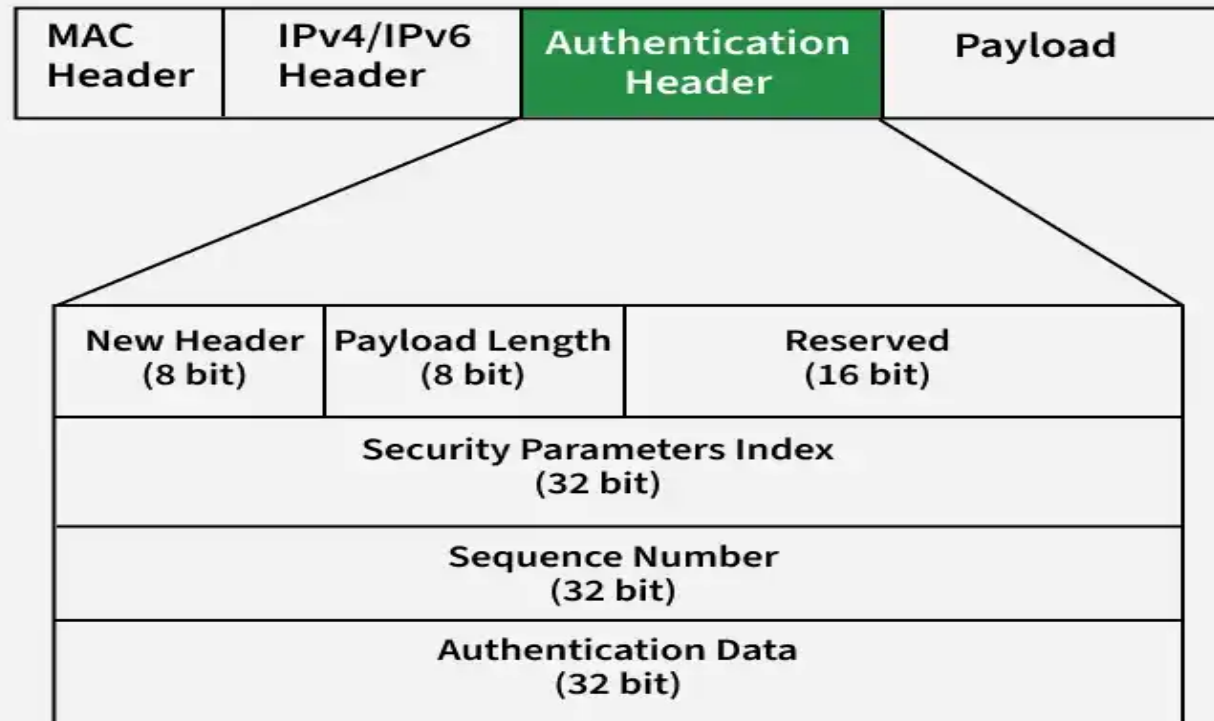
2. Transport Mode:

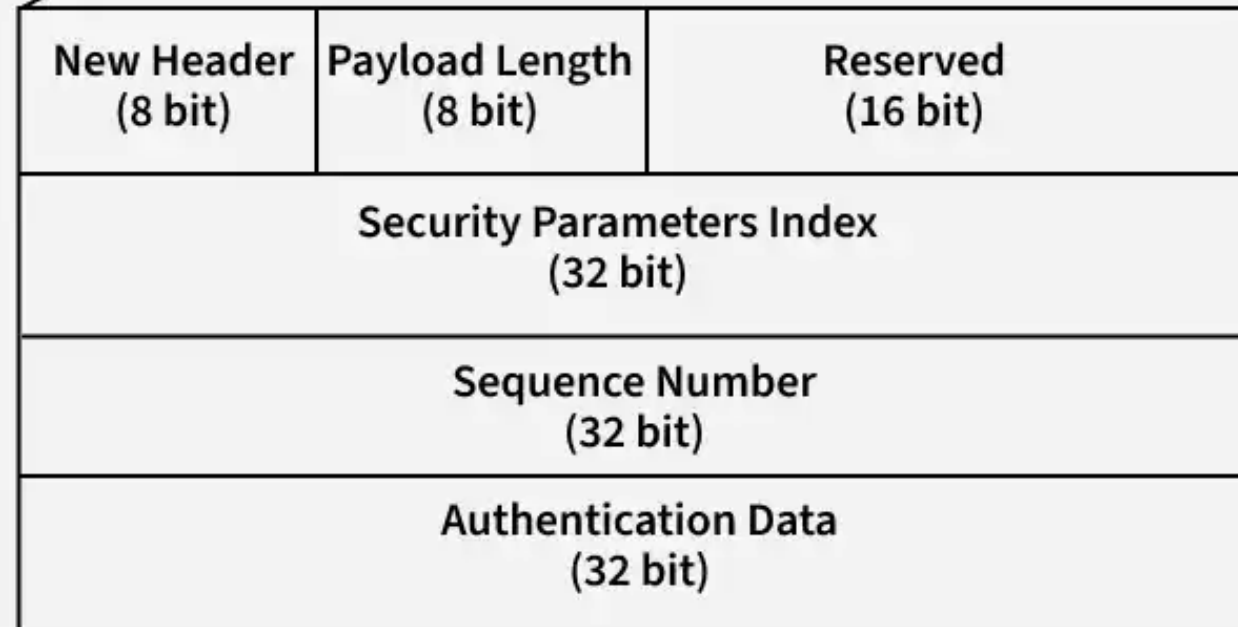
- In this mode, only the payload of the IP packet is encrypted, while the original IP header remains visible.
- Original IP header is not encrypted, so routing information remains exposed.
- Requires less processing compared to tunnel mode, resulting in faster communication.
- Commonly used for end-to-end communication between two hosts.

AH Protocol

Authentication Header (AH) Protocol

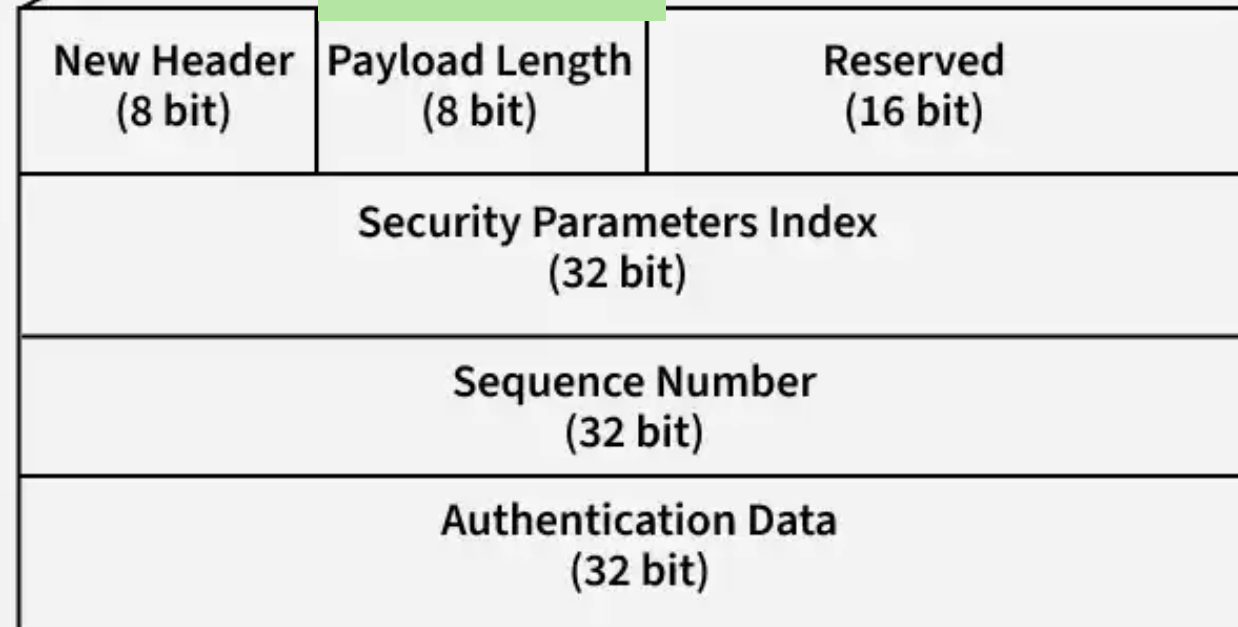
- AH Protocol provides both Authentication and Integrity service.
- Authentication Header is implemented in one way only: Authentication along with Integrity.

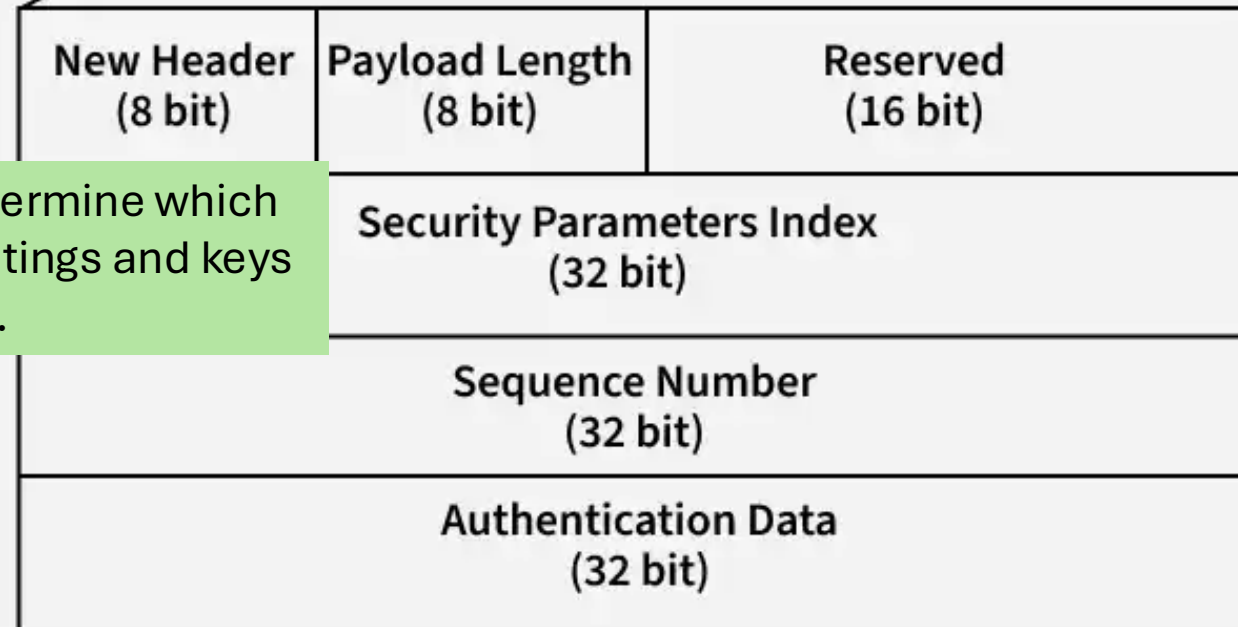




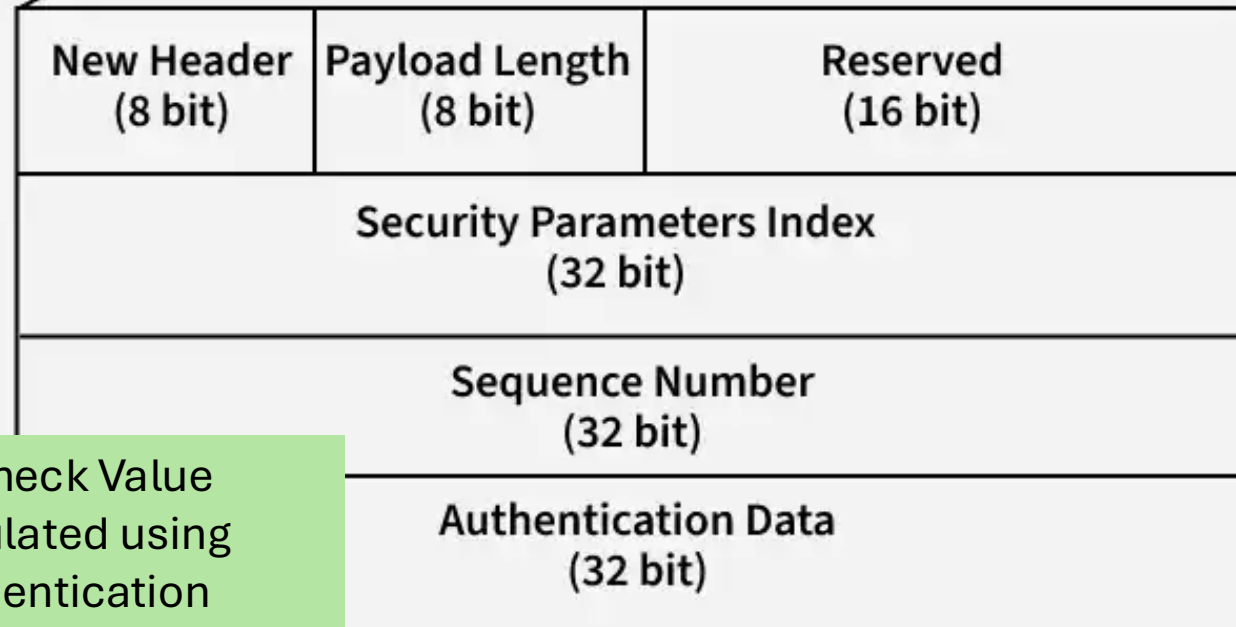


length of the Authentication Header.





Used to determine which security settings and keys are applied.



Integrity Check Value (ICV) calculated using keyed authentication algorithm such as HMAC.

1. Next Header (8 bits)

- Identifies the type of protocol (TCP, UDP, ICMP or another protocol) that comes after the AH header.

2. Payload Length (8 bits)

- Specifies the length of the Authentication Header.
- Measured in 32-bit words minus 2.
- Helps the receiver determine where the AH ends and the payload begins.

3. Reserved (16 bits)

- Reserved for future use. (Currently set to zero.)

4. Security Parameters Index (SPI) (32 bits)

- Identifies the Security Association (SA) between sender and receiver.
- Used to determine which security settings and keys are applied to the packet.

5. Sequence Number (32 bits)

- Used to prevent replay attacks by assigning a unique sequence number to every packet.

6. Authentication Data (variable length, multiple of 32 bits)

- Contains the Integrity Check Value (ICV).
- Generated using a cryptographic hash function.
- Ensures the packet has not been modified during transmission.

Modes of Operation

1. Transport Mode

- The **Authentication Header is inserted between the original IP header and the transport layer header (TCP/UDP).**
- Protects the payload and selected parts of the IP header
- **Original IP header remains visible**
- Used mainly for host-to-host communication
- Provides authentication and integrity, but no encryption

2. Tunnel Mode

- The **entire original IP packet (original IP header + data) is authenticated.** **A new outer IP header is added** and the Authentication Header is placed between the new IP header and the original IP packet.
- Protects the entire original packet
- Inner IP header contains the actual source and destination IP addresses
- Outer IP header contains IP addresses of security gateways (firewalls or VPN routers)
- **Commonly used in VPN communication**

DOI (Domain of Interpretation):

- DOI is the identifier that supports both AH and ESP protocols.
- Defines the common meaning of security parameters
- It contains values needed for documentation related to each other.

Key Management:

- Key Management contains the document that describes how the keys are exchanged between sender and receiver.
- It means creating, exchanging, storing, and managing the secret keys used for security.

Thank You !